Local Normal Forms for Complex Base Expansions: A General (s, b, l) Framework and the φ Specialization

VR & AI

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Notation crib (one page)

$l \geq 2$	sites per turn (angular radix)
$U = e^{i2\pi/l}$	unit rotation per site
$b > 1$	radial growth per site
$B = b\,U$	complex base
$\beta_l = b^l$	full-turn radial factor on one lattice
$P, Q \in \mathbb{Z}$	b satisfies $b^2 - Pb + Q = 0$ (quadratic)
U_n, V_n	Lucas sequences for (P, Q), $X_{n+1} = PX_n - QX_{n-1}$
$V_l = V_l(P, Q)$	per-track digit-set size
$k = r + lt$	track index ($r \in \{0, \ldots, l-1\}$, $t \in \mathbb{Z}$)
d_t	digit on a fixed track at site t
$\mathcal{D}$	target digit set (nonnegative or balanced)
$E_q,\ E_q^{\mathrm{lin}}$	weighted ℓ^1 and signed linear potentials

Abstract

We present a local, phase-aware normalization theory for complex base expansions built from three parameters: sites per turn $l \in \mathbb{N}_{\geq 2}$, per-step radial growth $b > 1$, and a per-level shrinkage factor $s \in (0, 1]$. For canonical integer-digit normalization we assume b is a quadratic algebraic integer. We prove a *tri-site, same-phase* carry identity that yields termination and confluence on each residue class ("phase track"), provide digit-set choices and carry-count bounds (with explicit denominator conditions), and show how to obtain global self-similarity with track-dependent angular lattices. We specialize to φ, recovering Lucas-number coefficients. We correct the termination proof to use the *signed* Q^l and provide an appendix covering the $Q^l < 0$ case.

1 Setup: the general (s, b, l) model

Fix integers $l \geq 2$ and parameters $b > 1$, $s \in (0, 1]$. Let

$$U := e^{i\,2\pi/l} \qquad \text{and} \qquad B := b\,U.$$

Definition 1.1 (Complex (b, l)-string; phase tracks)**.** A complex (b, l)-string is an integer sequence $(d_k)_{k \in \mathbb{Z}}$ satisfying:

1. *finite right tail:* $\exists K \in \mathbb{Z}$ such that $d_k = 0$ for all $k > K$;
2. *absolutely summable left tail:* $\sum_{t \geq 0} |d_{-t}|\, b^{-t} < \infty$.

Its value is the absolutely convergent series

$$\mathrm{Val}(d) := \sum_{k\in\mathbb{Z}} d_k\, B^k, \qquad \text{with } |B| = b.$$

Indices decompose into l *phase tracks* by residue $r \equiv k \pmod{l}$, where $k = r + lt$ and $t \in \mathbb{Z}$. On a fixed track r we write $d_t := d_{r+lt}$ and $B^{r+lt} = U^r\, b^r\, \beta_l^t$ with $\beta_l = b^l > 1$.

Definition 1.2 (Hierarchical sum with shrink s)**.** For levels $\ell = 0, 1, \ldots$, a finite-depth address chooses integers $d_{\ell,k}$ and positions k_ℓ; its value is

$$z \;=\; \sum_{\ell\geq 0} s^\ell \sum_{k\in\mathbb{Z}} d_{\ell,k}\, B^k.$$

For infinite depth, $s^\ell \to 0$ and absolute convergence of each inner sum ensure z is well-defined.

Remark 1.3 (Quadratic b for integer carries). To obtain *integer* carry coefficients (and thus integer digits under normalization), we assume b is a root of

$$b^2 - P\,b + Q \;=\; 0, \qquad P, Q \in \mathbb{Z}, \quad b > 1. \tag{1}$$

This includes all metallic means ($Q = -1$, $P \geq 1$) and their integer powers b^h.

Tracks and digit alphabets (notation)

Phase tracks. Indices decompose as $k = r + lt$ with $r \in \{0, \ldots, l-1\}$ and $t \in \mathbb{Z}$. On a fixed track r we write $d_t := d_{r+lt}$ and

$$B^{r+lt} = U^r\, b^r\, \beta_l^t, \qquad \beta_l := b^l > 1.$$

Digit alphabets. Throughout we normalize per track into a contiguous set $\mathcal{D}$ of size $V_l(P, Q)$:

$$\mathcal{D} = \begin{cases} \{0, 1, \ldots, V_l - 1\}, & \text{nonnegative,} \\ \{-a, \ldots, a\}, \quad a = \lfloor (V_l - 1)/2 \rfloor, & \text{balanced.} \end{cases}$$

Balanced digits are convenient when $Q^l < 0$ (odd l), as they avoid long borrow cascades; nonnegative digits are natural when $Q^l > 0$ (even l).

Convergence of hierarchical sums (sufficient bound)

Let $\beta_l := b^l > 1$. If, for each level ℓ, the per-track reindexed series obeys

$$\sum_{t\in\mathbb{Z}} |d_{\ell,r+lt}|\, \beta_l^t \;\leq\; C_\ell \quad \text{for all residues } r,$$

then

$$|z| \;\leq\; \sum_{\ell\geq 0} s^\ell \sum_{r=0}^{l-1} C_\ell.$$

Thus $s < 1$ and $\sup_\ell C_\ell < \infty$ imply absolute convergence.

Post-normalization quantification. After per-track normalization into a digit set of size $V := V_l(P, Q)$, each digit satisfies $|d_{\ell,r+lt}| \leq V - 1$ (or $\leq a = \lfloor (V-1)/2 \rfloor$ in the balanced case). If the normalized support on level ℓ has ν_ℓ nonzero sites in total and the maximal occupied site index is $t_{\max,\ell}$ (measured on its track), then the crude but effective bound

$$\boxed{C_\ell \;\leq\; (V-1)\, \beta_l^{t_{\max,\ell}}\, \nu_\ell}$$

holds, since $\beta_l^t \leq \beta_l^{t_{\max,\ell}}$ for all occupied t. When occupied sites form blocks, the sharper geometric sum $\sum_{t=a}^{b} \beta_l^t = \beta_l^a \frac{\beta_l^{b-a+1}-1}{\beta_l - 1}$ may be used trackwise; either way C_ℓ scales linearly in the (finite) active support at level ℓ.

Numerical illustration (φ, $l = 4$). Here $\beta_l = \varphi^4 \approx 6.854$. Suppose a level ℓ has $\nu_\ell = 3$ occupied sites on a track, at $t \in \{-1, 0, 1\}$, with digits bounded by $V - 1 = L_4 - 1 = 6$. The crude bound yields

$$C_\ell \;\le\; (V-1)\,\beta_l^{t_{\max,\ell}}\,\nu_\ell \;=\; 6 \cdot 6.854^{\,1} \cdot 3 \;\approx\; 123.4.$$

Using the geometric sum per block (here contiguous from -1 to 1):

$$\sum_{t=-1}^{1} \beta_l^t \;=\; \beta_l^{-1}\,\frac{\beta_l^3 - 1}{\beta_l - 1} \;\approx\; 0.146 \cdot \frac{321.996 - 1}{5.854} \;\approx\; 8.0,$$

so $C_\ell \le 6 \cdot 8.0 \approx 48$, a tighter bound.

2 Local identity and normalization on one lattice (b, l)

Let $U_n(P,Q), V_n(P,Q)$ be the Lucas sequences:

$$U_0 = 0,\, U_1 = 1, \quad V_0 = 2,\, V_1 = P, \quad X_{n+1} = PX_n - QX_{n-1}.$$

From (1) one has for $n \ge 1$:

$$b^{2n} = V_n(P,Q)\, b^n - Q^n. \tag{2}$$

Proposition 2.1 (Phase-aligned (tri-site) carry identity)**.** *For all $k \equiv r \pmod{l}$,*

$$B^{k+2l} - V_l(P,Q)\, B^{k+l} + Q^l\, B^k \;=\; 0. \tag{3}$$

Equivalently on a track (with $\beta_l = b^l > 1$):

$$\beta_l^{t+1} \;=\; V_l(P,Q)\,\beta_l^t \;-\; Q^l\,\beta_l^{t-1}.$$

Proof of Proposition 2.1. Let b be quadratic with $b^2 - Pb + Q = 0$, and let $U = e^{i2\pi/l}$. The Lucas identity $b^{2n} = V_n(P,Q)\, b^n - Q^n$ holds for all $n \ge 1$. Taking $n = l$ gives $b^{2l} = V_l(P,Q)\, b^l - Q^l$. Multiply by U^k and note $U^l = 1$:

$$(bU)^{k+2l} - V_l(P,Q)\,(bU)^{k+l} + Q^l\,(bU)^k \;=\; 0.$$

Since $B = bU$, this is exactly $B^{k+2l} - V_l(P,Q)\, B^{k+l} + Q^l B^k = 0$, and it holds for every $k \equiv r \pmod{l}$, i.e. along each residue track. □

Local moves (per track). For $t^\star \in \mathbb{N}$, define the value-preserving carry/borrow at center t by

$$\begin{aligned} \textbf{carry:}\ & (d_{t-1}, d_t, d_{t+1}) \mapsto (d_{t-1} + Q^l t^\star,\ d_t - V_l(P,Q)t^\star,\ d_{t+1} + t^\star), \\ \textbf{borrow:}\ & (d_{t-1}, d_t, d_{t+1}) \mapsto (d_{t-1} - Q^l t^\star,\ d_t + V_l(P,Q)t^\star,\ d_{t+1} - t^\star). \end{aligned} \tag{4}$$

Algorithm 1: Deterministic per-track normalization

Let $V := V_l(P,Q)$ and let the target digit set be either $\{0, \ldots, V-1\}$ (nonnegative) or the balanced set $\{-a, \ldots, a\}$ with $a = \lfloor (V-1)/2 \rfloor$. The following rightmost-first procedure yields the canonical normal form.

Input: integer array $(d_t)_{t \in \mathbb{Z}}$ with finite right tail. **Output:** normalized digits $(\hat{d}_t)$ in $\mathcal{D}$.

1. **Rightmost selection.** While some t has $d_t \notin \mathcal{D}$, choose the *maximum* such index $t^\star$.

2. **One-shot fix at $t^\star$.**

   - *Nonnegative digits:* If $d_{t^\star} \geq V$, set $t = \lfloor d_{t^\star}/V \rfloor$ and apply a carry with multiplicity t:
   $$(d_{t^\star-1}, d_{t^\star}, d_{t^\star+1}) \leftarrow (d_{t^\star-1} + Q^l t,\ d_{t^\star} - Vt,\ d_{t^\star+1} + t).$$
   If $d_{t^\star} < 0$, set $t = \lceil (-d_{t^\star})/V \rceil$ and apply a borrow of multiplicity t:
   $$(d_{t^\star-1}, d_{t^\star}, d_{t^\star+1}) \leftarrow (d_{t^\star-1} - Q^l t,\ d_{t^\star} + Vt,\ d_{t^\star+1} - t).$$
   - *Balanced digits:* Replace the division by V with centered rounding $t = \lfloor (d_{t^\star} + a)/V \rfloor$ so that the new center lands in $\{-a, \ldots, a\}$ in one step.

3. **Repeat** Step 1.

Remark (Sparse / NAF-like variants). If a sparser normal form is desired (e.g., forbidding adjacent nonzeros on a track, or minimizing a local adjacency cost), two light-touch adaptations keep the theory intact: (i) a rightmost-first selection with a tie-breaker that reduces the adjacency count locally; and (ii) a bounded post-normalization sparsify pass (balanced digits) that removes adjacent pairs via a single local carry/borrow while keeping digits in $\mathcal{D}$. Both preserve termination and confluence.

Theorem 2.2 (Termination and confluence (single lattice))**.** *Fix a track and $q \in (1, \beta_l)$ with $\beta_l = b^l$. With the* signed *linear potential $E_q^{\text{lin}} = \sum_t d_t q^t$, a unit carry at t changes E_q^{lin} by $q^{t-1}(q^2 - V_l q + Q^l)$, which is negative for all $q \in (1, \beta_l)$. Hence normalization terminates on the track. Overlaps of carries occur only at adjacent centers; critical pairs are joinable. Summing the l tracks yields a unique normal form.*

Termination. A unit carry at t changes (d_{t-1}, d_t, d_{t+1}) by $(+Q^l, -V_l, +1)$, hence $\Delta E_q^{\text{lin}} = q^{t-1}(q^2 - V_l q + Q^l) = (q - \beta_l)(q - Q^l/\beta_l)\, q^{t-1} < 0$ for $q \in (1, \beta_l)$ since $Q^l/\beta_l < 1 < \beta_l$. For the nonnegative target with $Q^l > 0$ (e.g. metallic means and even l), all steps are carries so $E_q^{\text{lin}} \geq 0$ strictly decreases. For $Q^l < 0$ see Appendix A. □

Lemma 2.3 (Critical-pair check (adjacent carries commute))**.** *Let C_t denote a unit carry at center t (i.e. $t^\star = 1$), and write $V := V_l(P, Q)$, $Q_0 := Q^l$. Restrict to $\{t-1, t, t+1, t+2\}$ with digits (a, b, c, d). Then*
$$C_{t+1}\, C_t(a, b, c, d) \;=\; C_t\, C_{t+1}(a, b, c, d) \;=\; (a + Q_0,\ b - V + Q_0,\ c + 1 - V,\ d + 1).$$
By linearity, carries with $t^\star > 1$ decompose into sums of unit carries. Hence overlapping carries at t and $t+1$ commute.

Proof. Apply the updates in both orders and compare; the two linear maps coincide on (a, b, c, d). □

Confluence. Local confluence follows from commutation of adjacent carries and disjointness of nonadjacent ones. With termination, Newman's lemma yields global confluence and a unique per-track normal form. □

3 Digit-set variants and fast addition

- **Balanced signed digits.** With $\mathcal{D} = \{-a, \ldots, a\}$, a provisional sum s_t satisfies $|s_t| \leq 2a \leq V_l - 1$. A *single* carry at t reduces $|s_t|$ to $\leq a$; this yields one-pass, bounded-carry addition per track (plus bounded neighbor fixes).
- **Sparse (NAF-like).** Use a small signed set (e.g. $\{-1, 0, +1\}$) and forbid adjacent nonzeros on a track. Add a local penalty to the potential to keep termination; sparsity reduces Hamming weight.

4 Carry-count bounds (per track)

All bounds below presuppose the denominators are positive. In particular, the uniform ℓ^1 estimate requires $V_l(P,Q) > (|Q|^l + 1)$; when this gap is small, prefer the weighted bound.

Proposition 4.1 (Uniform ℓ^1 bound)**.** *Assume $V_l(P,Q) > (|Q|^l + 1)$. Each carry reduces the center by $V_l(P,Q)$ and increases the two neighbors' absolute sum by at most $|Q|^l + 1$. Hence*

$$\#\text{carries} \;\le\; \frac{S}{V_l(P,Q) - (|Q|^l + 1)}.$$

Proposition 4.2 (Weighted bound)**.** *For $q \in (1, b^l)$ and $E_q = \sum_t |d_t| q^t$, if $V_l(P,Q)^2/4 - |Q|^l > 0$ (e.g. $q = V_l/2 < b^l$), then*

$$\#\text{carries} \;\le\; \frac{E_q^{\text{init}}}{\frac{V_l(P,Q)^2}{4} - |Q|^l}.$$

Carry-count instantiation (φ, $l = 4$). On one track with initial digits $(0, 13, 12, 0)$ we have $S = 25$, $V = 7$, $|Q|^l = 1$. The uniform ℓ^1 bound gives $\# \le 25/(7-2) = 5$. With $q = 3.5$, $E_q = 13 + 12q = 55$, so the weighted bound gives $\# \le 55/(49/4 - 1) \approx 4.89$. A concrete normalization needs exactly 4 carries, showing both bounds are nontrivial and reasonably tight.

Bound tightness across configurations

We compare the uniform ℓ^1 bound and the weighted bound on several initial states for $(\varphi, l{=}4)$, using rightmost-first normalization with unit-carry counting.[1]

case	S	E_q (q=3.5)	uniform	weighted	actual #
A sparse center $(0, 13, 0)$	13	13.000	2.600	1.156	1
B adjacent heavy $(0, 13, 12, 0)$	25	55.000	5.000	4.889	4
C dense block $(12, 12, 12, 12, 12)$	60	205.408	12.000	18.259	10
D far-right spike $(0, 0, 0, 0, 50)$	50	612.500	10.000	54.444	9

5 Track-dependent l_r with global self-similarity

Fix a *global similarity exponent* $M \in \mathbb{N}$ and set $\beta := b^M > 1$. Choose tracks r; for each track pick $l_r \mid M$ and define

$$h_r := \frac{M}{l_r} \in \mathbb{N}, \qquad B_r := b^{h_r}\, e^{i2\pi/l_r}.$$

Then $(b^{h_r})^{l_r} = \beta$ (same full-turn radial factor on all tracks). With level shrink s^ℓ, the value is

$$z = \sum_{\ell \ge 0} s^\ell \sum_r \sum_{k \in \mathbb{Z}} d_{r,\ell,k}\, B_r^k.$$

Proposition 5.1 (Uniform carry across tracks)**.** *If b satisfies $x^2 - Px + Q = 0$, then for each track r and $k \equiv r \pmod{l_r}$,*

$$B_r^{k+2l_r} - V_M(P,Q)\, B_r^{k+l_r} + Q^M\, B_r^k = 0. \tag{5}$$

Thus every track uses the same *coefficients $V_M(P,Q)$ and Q^M; the digit-set size is $V_M(P,Q)$. Termination/confluence and bounds hold uniformly with $\beta = b^M$.*

[1] Actual carry counts were computed by Algorithm 1, counting a carry of multiplicity t as t unit tri-site carries.

Proof of Proposition 5.1. Fix the global $M \in \mathbb{N}$ and a track with $l_r \mid M$, $h_r = M/l_r$. Then b^{h_r} is quadratic over $\mathbb{Z}$ with the same (P, Q), hence

$$(b^{h_r})^{2l_r} = V_{l_r}(P, Q)\,(b^{h_r})^{l_r} - Q^{l_r}.$$

Because $h_r l_r = M$, the composed Lucas coefficient is $V_{h_r l_r}(P, Q) = V_M(P, Q)$ and $Q^{h_r l_r} = Q^M$. Thus $b^{2M} = V_M(P, Q)\, b^M - Q^M$. Multiplying by U_r^k with $U_r^{l_r} = 1$ yields

$$B_r^{k+2l_r} - V_M(P, Q)\, B_r^{k+l_r} + Q^M B_r^k = 0,$$

which is track-independent. □

6 Specialization to φ

Set $b = \varphi = \frac{1+\sqrt{5}}{2}$, which satisfies $x^2 - x - 1 = 0$ ($P = 1$, $Q = -1$). Then

$$V_n(P, Q) = L_n \quad \text{(Lucas numbers)}, \qquad Q^n = (-1)^n.$$

Single lattice (φ, l)**.** The carry identity becomes

$$B^{k+2l} - L_l\, B^{k+l} + (-1)^l B^k = 0, \qquad B = \varphi\, e^{i2\pi/l},$$

and per-track digits normalize into a set of size L_l; with l even, $\{0, \ldots, L_l - 1\}$ is natural. The potential window is $(1, \varphi^l)$.

Example 6.1 (φ and quarter-turns). Take $l = 4$. Then $L_4 = 7$ and $B^{k+8} - 7B^{k+4} + B^k = 0$ ($(-1)^4 = +1$). Nonzeros, e.g. at $k \in \{-2, 2, 5\}$, decompose by residues: $r = 2$ has $t = -1$ and $t = 0$; $r = 1$ has $t = 1$; tracks $r = 0, 3$ empty. Since digits lie in $\{0, \ldots, 6\}$, no carries are needed here.

7 Figures (selected)

8 Related work (brief pointers)

Complex bases and non-standard radices have a rich history. Knuth [6] discusses negative and unusual bases in a foundational algorithmic context. Penney [7] introduced base $-1 + i$ arithmetic. Gilbert [3, 4] studied arithmetic and representations in complex bases. Frougny and Sakarovitch [2] surveyed automatic/normalized representations in algebraic bases. Our contribution is a *phase-aware tri-site* normalization that (i) works uniformly on residue tracks, (ii) gives simple termination/confluence proofs via a potential, and (iii) extends cleanly to track-dependent lattices with *uniform* carry coefficients under a global self-similarity constraint (see also [5, 1] for modern developments on canonical number systems in quadratic/imaginary settings).

Appendix A: Termination for $Q^l < 0$ (sketch)

Two variants keep the Lyapunov argument valid for odd l (e.g. metallic means).

1. **Balanced digits (macro-steps).** Normalize into $\mathcal{D} = \{-a, \ldots, a\}$. Group updates into macro-steps: perform a rightmost carry at t (unit multiplicity) and, if d_{t-1} leaves $\mathcal{D}$, immediately fix it by at most one local borrow. The signed linear potential E_q^{lin} decreases by $q^{t-1}(q^2 - V_l q + Q^l)$ per macro-step for any $q \in (1, \beta_l)$.

2. **Two-sided Lyapunov.** Use $E_q^\lambda = E_q^{\text{lin}} + \lambda \sum_t |d_t| q^t$ with small $\lambda > 0$. Carries decrease E_q^{lin} by a fixed amount; transient borrows change the ℓ^1 term by $O(q^{t-1})$ with a controlled coefficient. Choose λ so the net change is negative.

Local confluence remains identical (adjacent tri-site updates commute), so Newman's lemma yields uniqueness.

Appendix B: Tiny reference implementation (for $\varphi, l = 4$)

A 30-line Python script (rightmost-first, nonnegative digits $\{0, \ldots, 6\}$) for per-track normalization and bound checks is provided as supplementary material (`phi_l4_reference.py`). Core loop:

```
V = 7; Qpow = 1
def normalize_rightmost_first(d):
    d = d.copy(); unit_carries = 0
    while True:
        cand = [t for t,v in d.items() if v >= V]
        if not cand: break
        tstar = max(cand); tmult = d[tstar] // V
        d[tstar-1] = d.get(tstar-1, 0) + Qpow * tmult
        d[tstar]   = d[tstar] - V * tmult
        d[tstar+1] = d.get(tstar+1, 0) + tmult
        unit_carries += tmult
    return unit_carries, d
```

References

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